

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T37 · Transformations of Graphs

With Answers

Questions with full worked solutions

■ 52 pages

FAMILY MAP

Transformations of Graphs

F01	Map a known graph feature	17 Q 28 marks
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F02	Infer or describe the graph transformation	3 Q 7 marks
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F03	Write the transformed equation	5 Q 10 marks
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19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are $(8, -12)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(.....,)
(1)

(ii) $y = f(2x)$

(.....,)
(1)

(Total for Question 19 is 2 marks)

Worked Solution 12

Transformations of a minimum point

UNIT 2**2 marks****Vertical translation**

$$y = f(x) + 3 : \quad (8, -12 + 3) = (8, -9).$$

Horizontal scaling

$$y = f(2x) : \quad 2x = 8 \Rightarrow x = 4, \text{ so } (4, -12).$$

Final answer

$$(8, -9), \quad (4, -12)$$

Question 09

4MA1-1H Q21

ORIGINAL QUESTION

2 marks

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(5, -4)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(..... ,)
(1)

(ii) $y = f(x) - 6$

(..... ,)
(1)

Worked Solution 09

Graph transformations

MODULAR UNIT 2

2 marks

Horizontal shift

$$y = f(x + 7) : (5, -4) \mapsto (-2, -4).$$

Vertical shift

$$y = f(x) - 6 : (5, -4) \mapsto (5, -10).$$

Final answer

$$(-2, -4); \quad (5, -10)$$

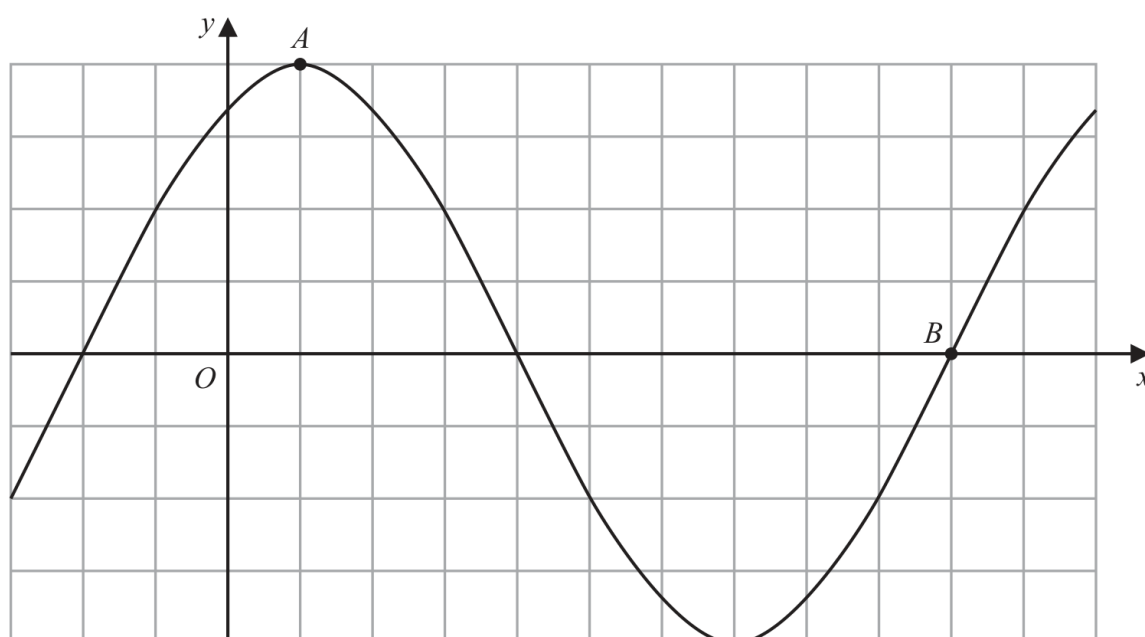
Question 27

4MA1-2H Q25

ORIGINAL QUESTION

2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

Worked Solution 27

Transformed sine graph

MODULAR UNIT 2

2 marks

Maximum

$$x + 60 = 90 \Rightarrow A = (30, 2).$$

Axis crossing

$$x + 60 = 360 \Rightarrow B = (300, 0).$$

Final answer

$$A = (30, 2), \quad B = (300, 0)$$

Question 19

4MA1-1H Q24 (a)

ORIGINAL QUESTION UNIT 2

2 marks

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

(a) Write down the value of a and the value of b

 $a = \dots\dots\dots$ $b = \dots\dots\dots$

(2)

Worked solution 19

Translation of a function

MODULAR UNIT 2

2 marks

(a) – Translation of a function

Match the x shift

 $g(x) = f(x + 4) + 6$, so the graph moves 4 left and 6 up.**Final answer** $(a,b) = (-4, 6)$

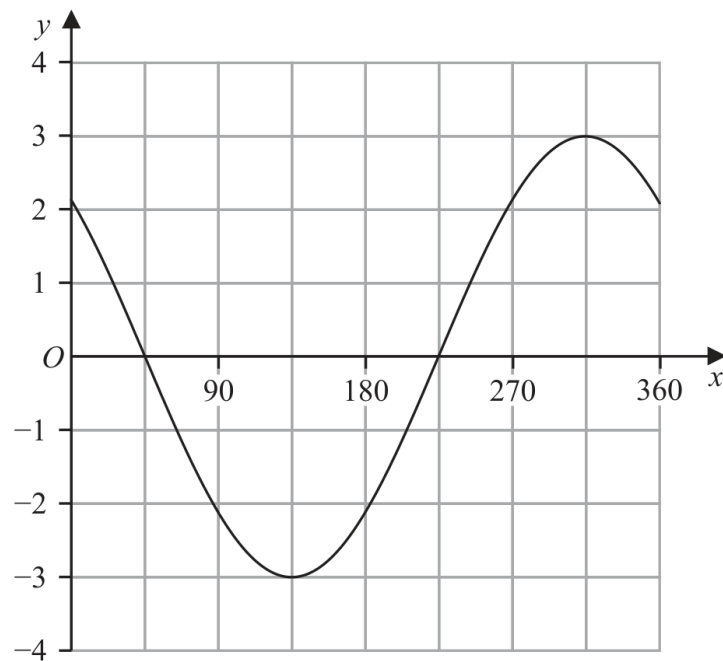
Question 20

4MA1-1H Q24 (b)

ORIGINAL QUESTION UNIT 2

2 marks

The graph of $y = p \cos(x + q)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(2)

Worked solution 20Read p and q from a cosine graph

MODULAR UNIT 2

2 marks

(b) – Read p and q from a cosine graph

Read the amplitude

$$\max y = 3, \quad p = 3 \ (p > 0).$$

Read the phase shift

$$\text{The maximum occurs at } x = 315^\circ; \ 315 + q = 360 \Rightarrow q = 45^\circ.$$

Final answer

$$p = 3, \ q = 45$$

Question 17

4MA1-1H Q21

ORIGINAL QUESTION

2 marks

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.....,)
(1)

(b) $y = f(3x)$

(.....,)
(1)

Worked Solution 17

Turning point under graph transformations

MODULAR UNIT 2

2 marks

Horizontal translation

$$y = f(x - 4) : (6, 5) \mapsto (10, 5).$$

Horizontal scale

$$y = f(3x) : (6, 5) \mapsto (2, 5).$$

Final answer

$$(10, 5); \quad (2, 5)$$

Original question 30

4MA1-2H Q19 M01

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 5)$

(.....,)
(1)

Worked solution 30

Translate a graph horizontally

MODULAR UNIT 2

1 marks

M01 – Translate a graph horizontally**Track x-values** $y=f(x+5)$ shifts the minimum (5,4) left by 5, giving (0,4).**Final answer**

(0,4)

Original question 31

4MA1-2H Q19 M02

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

(ii) $y = 3f(x)$

(.....,)
(1)

Worked solution 31

Stretch graph values

MODULAR UNIT 2

1 marks

M02 – Stretch graph values**Multiply the output**

For $y=3f(x)$, the x-coordinate stays 5 and the y-coordinate becomes $3(4)=12$.

Final answer

(5,12)

Original question 32

4MA1-2H Q19 M03

MODULAR UNIT 2

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

(iii) $y = f(x) - 7$

(.....,)
(1)

Worked solution 32

Shift a graph vertically

MODULAR UNIT 2

1 marks

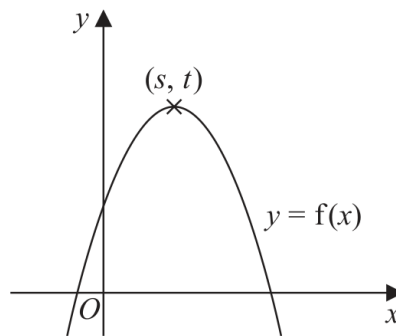
M03 – Shift a graph vertically**Subtract 7 from y**For $y=f(x)-7$, $(5,4)$ becomes $(5,4-7)=(5,-3)$.**Final answer** $(5,-3)$

Original question 16

4MA1-1H Q20 Q20(i)

ORIGINAL QUESTION**1 marks**

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (s, t)

Find, in terms of s and t , the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 2)$

(.....,)
(1)

Worked solution 16

Chapter: Transformations of Graphs

MODULAR UNIT 2

1 marks

Q20(i) – Chapter: Transformations of Graphs

Family: Map a known graph feature • Variant: Horizontal translation

Method

Replacing x by $x - 2$ translates the graph 2 units right, so the maximum moves from (s, t) to $(s + 2, t)$.

Why: For $y = f(x - a)$, every point shifts a units in the positive x -direction.

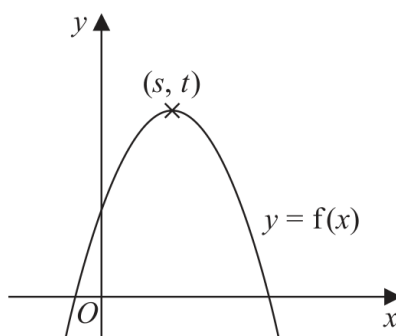
Final answer**maximum:** $(s + 2, t)$

Original question 17

4MA1-1H Q20 Q20(ii)

ORIGINAL QUESTION**1 marks**

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (s, t)

Find, in terms of s and t , the coordinates of the maximum point on the curve with equation

(ii) $y = 3f(x)$

(.....,)

(1)

Worked solution 17

Chapter: Transformations of Graphs

MODULAR UNIT 2

1 marks

Q20(ii) – Chapter: Transformations of Graphs

Family: Map a known graph feature • Variant: Vertical scale or x-axis reflection

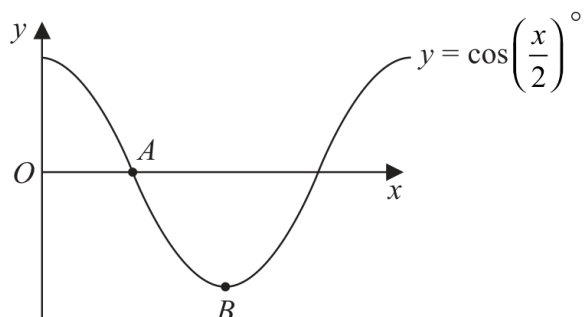
MethodMultiplying $f(x)$ by 3 keeps the x-coordinate and triples the y-coordinate, so the maximum is $(s, 3t)$.*Why: The transformation $y = 3f(x)$ is a vertical stretch by scale factor 3.***Final answer****maximum:** $(s, 3t)$

Original question 31

4MA1-2H Q23 Q23(i)

ORIGINAL QUESTION**1 marks**

- 23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



- (i) Find the coordinates of the point A

(.....,)
(1)

Worked solution 31

Chapter: Transformations of Graphs

MODULAR UNIT 2

1 marks

Q23(i) – Chapter: Transformations of Graphs

Family: Map a known graph feature • Variant: Horizontal scale or y-axis reflection

Method

At A, $\cos(x/2) = 0$. The first positive solution is $x/2 = 90$ degrees, so $A = (180, 0)$.

Why: Read the first positive x-intercept of the cosine graph in degree mode.

Final answer

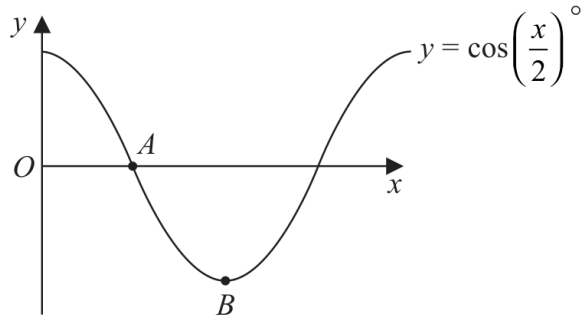
A: $A = (180, 0)$

Original question 32

4MA1-2H Q23 Q23(ii)

ORIGINAL QUESTION**1 marks**

- 23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



- (ii) Find the coordinates of the point B

(.....,)
(1)

Worked solution 32

Chapter: Transformations of Graphs

MODULAR UNIT 2

1 marks

Q23(ii) – Chapter: Transformations of Graphs

Family: Map a known graph feature • Variant: Horizontal scale or y-axis reflection

Method

At the first minimum, $x/2 = 180$ degrees, so $x = 360$ degrees and $y = -1$; hence $B = (360, -1)$.

Why: Use the first minimum of the cosine graph in degree mode.

Final answer

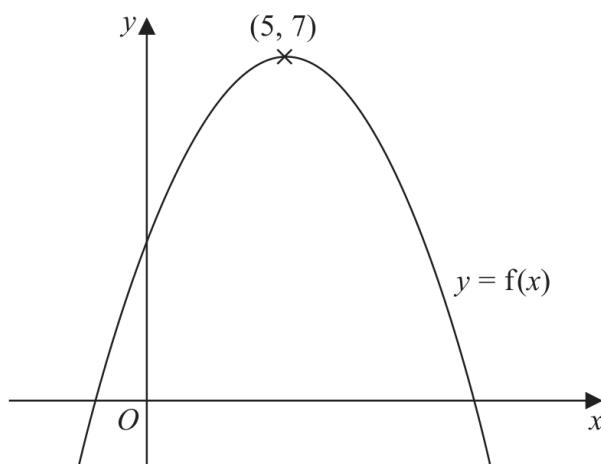
B: $B = (360, -1)$

Original question 14

4MA1-1H Q23

ORIGINAL QUESTION**2 marks**

- 23 The diagram shows a sketch of the curve with equation $y = f(x)$



There is only one maximum point on the curve.
The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

Worked solution 14

Chapter: Transformations of Graphs

MODULAR UNIT 2

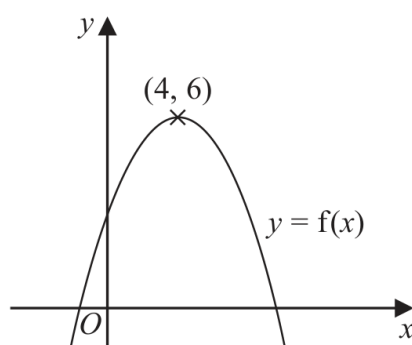
2 marks

Q23 – Chapter: Transformations of Graphs

Family: Map a known graph feature • Variant: Horizontal translation

MethodFor $y=f(x+9)$, the graph moves 9 units left, so $(5,7)$ becomes $(-4,7)$.*Why: An input translation $x+9$ shifts the graph left by 9.***Method**For $y=f(x)+3$, the graph moves 3 units up, so $(5,7)$ becomes $(5,10)$.*Why: An output translation $+3$ shifts all y -values up by 3.***Final answer**maximum for $y=f(x+9)$: $(-4,7)$ **Final answer**maximum for $y=f(x)+3$: $(5,10)$

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.....,)

(ii) $y = f(2x)$

(.....,)

(2)

Worked solution

January 2020 | Question 21 | 2 marks

Family: Map a known graph feature • Variant: Combined feature mapping

Method / working

1. $f(x+4)$ shifts the maximum 4 left: $(4,6) \rightarrow (0,6)$.

2. $f(2x)$ halves the x-coordinate: $(4,6) \rightarrow (2,6)$.

Final answer: $(0,6)$ and $(2,6)$

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

.....
(2)

Worked solution

January 2020 | Question 21 | 2 marks

Family: Write the transformed equation • Variant: Transform an explicit polynomial and simplify

Method / working

1. Translate 4 right: replace x by $x-4$.

2. Translate 6 up: $y=(x-4)^2+3(x-4)+4+6$.

Final answer: $y=(x-4)^2+3(x-4)+10$

21 A curve has equation $y = f(x)$

The coordinates of the minimum point on this curve are $(-9, 15)$

(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 3)$

(.....,)

(ii) $y = \frac{1}{3} f(x)$

(.....,)

(2)

Worked solution

January 2021 | Question 21 | 2 marks

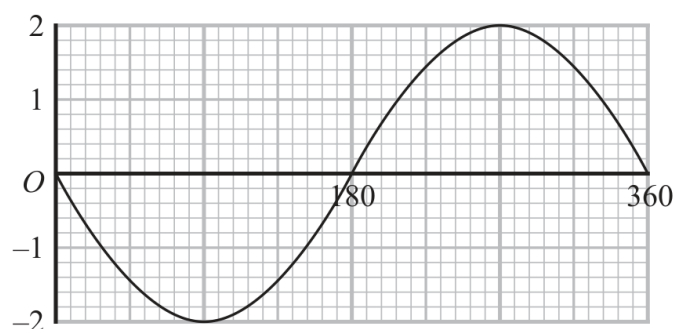
Family: Map a known graph feature • Variant: Combined feature mapping

Method / working

1. Replacing x by $x+3$ shifts the graph 3 units left.
2. The minimum point becomes $(-12,15)$.

Final answer: $(-12,15)$

The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $a > 0$ and that $0 < b < 360$

(b) find the value of a and the value of b .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

Worked solution

January 2021 | Question 21 | 2 marks

Family: Map a known graph feature • Variant: Combined feature mapping

Method / working

1. Multiplying $f(x)$ by $1/3$ scales all y-coordinates by $1/3$.
2. The minimum point becomes $(-9,5)$.

Final answer: $(-9,5)$

- 22** The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants,
describe fully a series of transformations that map the curve **S** onto the curve **T**.

.....

.....

.....

Worked solution

June 2021 | Question 22 | 4 marks

Family: Infer or describe the graph transformation | Variant: Describe from two curve equations

METHOD / WORKING

1. Completing the square: $2x^2 - 12x + 13 = 2(x - 3)^2 - 5$.2. The factor 2 is a vertical stretch of $y = x^2$.

3. Translate the vertex 3 units right.

4. Translate 5 units down; state $g(x)$.

FINAL ANSWER

$$g(x) = 2(x - 3)^2 - 5$$

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

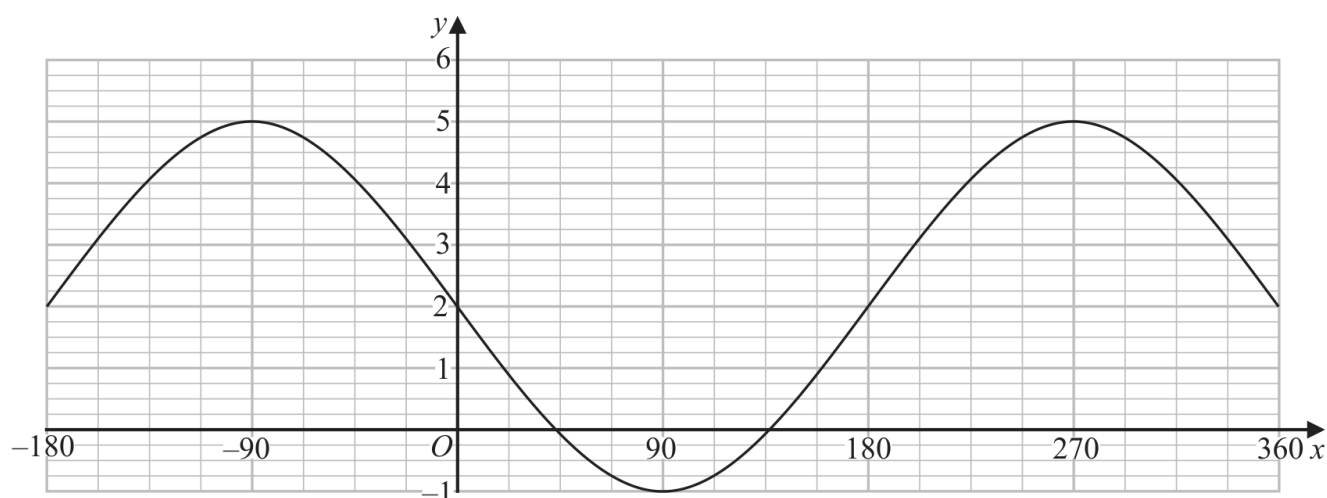
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(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Write the transformed equation | Variant: Transform an explicit polynomial and simplify

METHOD / WORKING

1. $f(x)=9-3(x+2)^2$

2. This is in completed square form, so the maximum point is

3. $(-2,9)$

4. For the translation

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

6. the graph moves 4 units right. Replace x by $x - 4$:

7. $y=9-3((x-4)+2)^2$

8. $y=9-3(x-2)^2$

9. The curve T has equation

10. $y=3(x+2)^2-9$

11. This is

12. $y=-\left(9-3(x+2)^2\right)$

13. So C is reflected in the x -axis.14. For the cosine graph, the maximum is 5 and the minimum is -1 .

15. $a = \frac{5 - (-1)}{2} = 3$

16. $c = \frac{5 + (-1)}{2} = 2$

17. For

18. $y=a\cos(x-b)^\circ+c$

19. a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

20. $b=270$

21. Answer: $A = (-2, 9)$, $y = 9 - 3(x - 2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

FINAL ANSWER

 $A=(-2,9)$, $y=9-3(x-2)^2$, reflection in the x -axis, $a=3$, $b=270$, $c=2$

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The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

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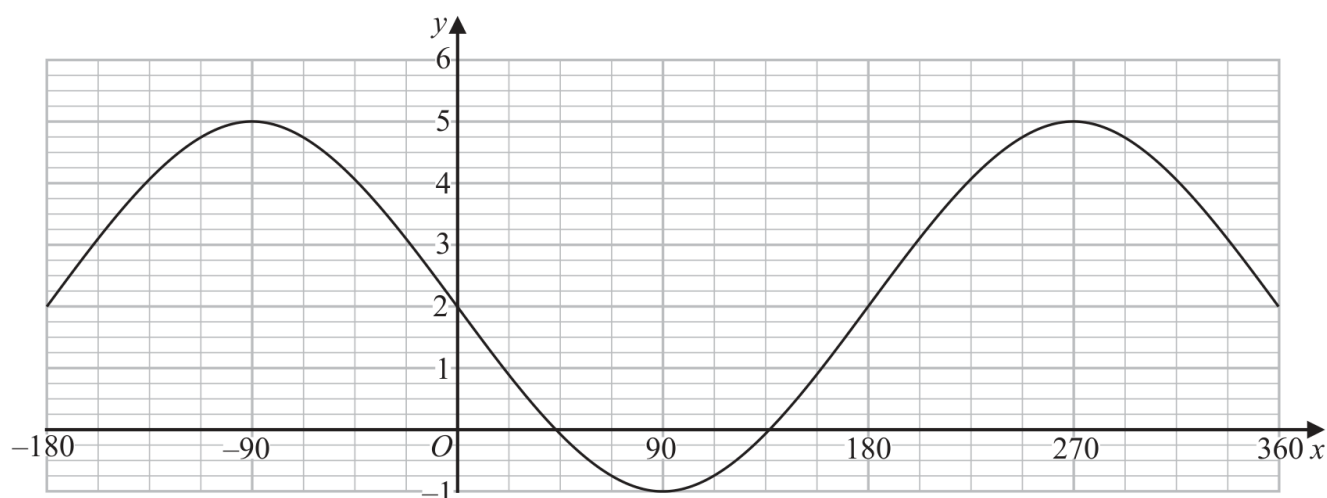
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(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Write the transformed equation | Variant: Translation vector applied to $y=f(x)$

METHOD / WORKING

1. $f(x)=9-3(x+2)^2$

2. This is in completed square form, so the maximum point is

3. $(-2,9)$

4. For the translation

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

6. the graph moves 4 units right. Replace x by $x - 4$:

7. $y=9-3((x-4)+2)^2$

8. $y=9-3(x-2)^2$

9. The curve T has equation

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17. For

18. $y=a\cos(x-b)^\circ+c$

19. a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

20. $b=270$

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FINAL ANSWER

 $A=(-2,9)$, $y=9-3(x-2)^2$, reflection in the x -axis, $a=3$, $b=270$, $c=2$

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

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(b) Find an equation for the curve **S**.

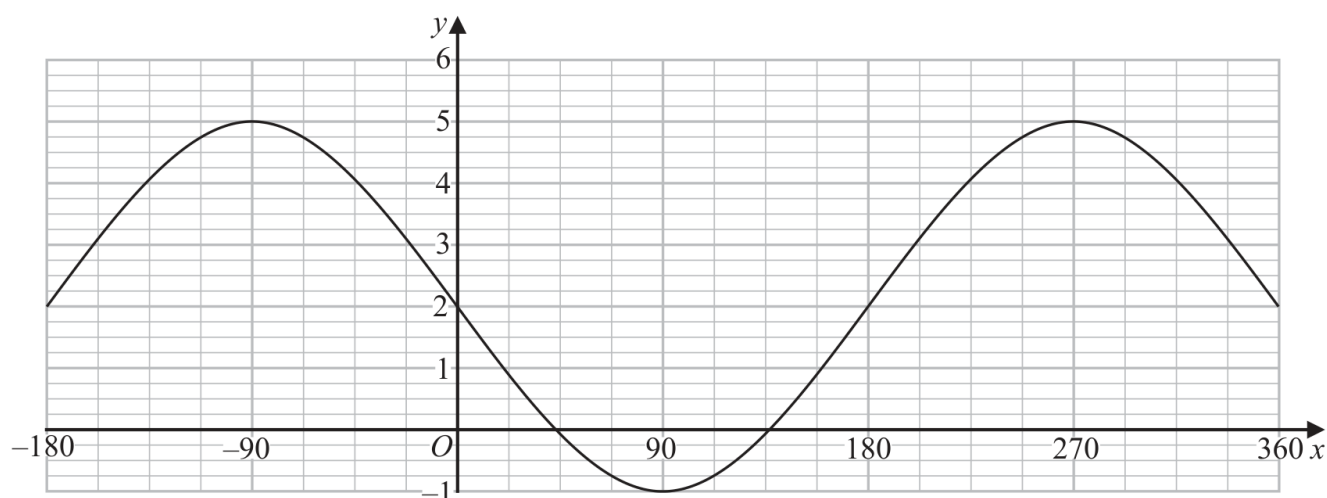
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Infer or describe the graph transformation | Variant: Describe from two curve equations

METHOD / WORKING

1. $f(x)=9-3(x+2)^2$

2. This is in completed square form, so the maximum point is

3. $(-2,9)$

4. For the translation

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

6. the graph moves 4 units right. Replace x by $x - 4$:

7. $y=9-3((x-4)+2)^2$

8. $y=9-3(x-2)^2$

9. The curve T has equation

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18. $y=a\cos(x-b)^\circ+c$

19. a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

20. $b=270$

21. Answer: $A = (-2, 9)$, $y = 9 - 3(x - 2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

FINAL ANSWER

 $A=(-2,9)$, $y=9-3(x-2)^2$, reflection in the x -axis, $a=3$, $b=270$, $c=2$

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(b) Find an equation for the curve **S**.

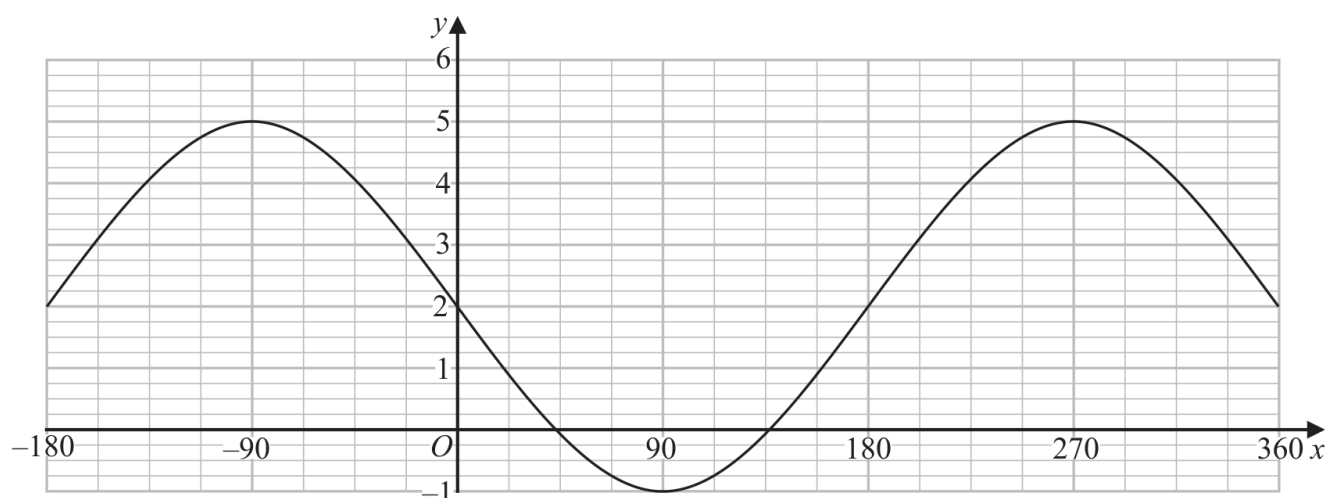
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 3 marks

Family: Map a known graph feature | Variant: Combined feature mapping

METHOD / WORKING

1. $f(x)=9-3(x+2)^2$

2. This is in completed square form, so the maximum point is

3. $(-2,9)$

4. For the translation

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

6. the graph moves 4 units right. Replace x by $x - 4$:

7. $y=9-3((x-4)+2)^2$

8. $y=9-3(x-2)^2$

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16. $c = \frac{5 + (-1)}{2} = 2$

17. For

18. $y=a\cos(x-b)^\circ+c$

19. a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

20. $b=270$

21. Answer: $A = (-2, 9)$, $y = 9 - 3(x - 2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

FINAL ANSWER

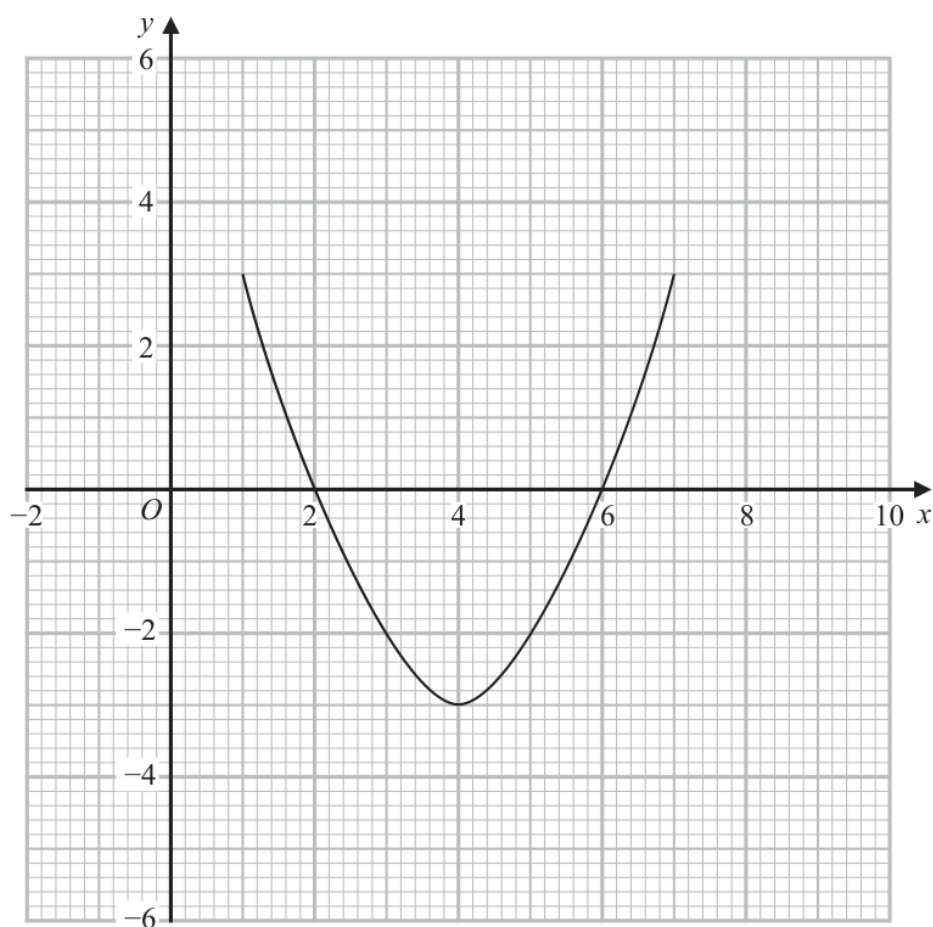
 $A=(-2,9)$, $y=9-3(x-2)^2$, reflection in the x -axis, $a=3$, $b=270$, $c=2$

- 25** The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

Answer reference | May-Nov 2020 | Question 25

Family: Write the transformed equation • Variant: Transform an explicit polynomial and simplify

25	(a)		Reflection in $y = 0$	1	B1 accept alternative for $y = 0$ e.g. x axis ; if more than one transformation then B0
	(b)		U shaped curve through (2, 6) (3, 0) (5, -6) (7, 0) (8, 6)	2	B2 for a U shaped curve passing through (2, 6) (3, 0) (5, -6) (7, 0) (8, 6) If not B2 then award B1 for either $2f(x - 1)$ passing through at least 3 points from (2, 6) (3, 0) (5, -6) (7, 0) (8, 6) or $2f(x + 1)$ passing through (0, 6) (1, 0) (3, -6) (5, 0) (6, 6) or $2f(x)$ passing through all of (1, 6) (2, 0) (4, -6) (6, 0) (7, 6) or $f(x - 1)$ passing through all of (2, 3) (3, 0) (5, -3) (7, 0) (8, 3) or $2f(x \pm k)$ passing through all of $(1 \pm k, 6)$ $(2 \pm k, 0)$ $(4 \pm k, -6)$ $(6 \pm k, 0)$ $(7 \pm k, 6)$ or A clear translation of the curve using the vector $\begin{pmatrix} 1 \\ k \end{pmatrix}$
Total 3 marks					

Worked solution

May-Nov 2020 | Question 25 | 3 marks

Family: Write the transformed equation • Variant: Transform an explicit polynomial and simplify

Method

For part (a),

$y=g(x)$ to $y=-g(x)$

The x-coordinates stay the same and the y-coordinates change sign.

So the transformation is a reflection in the x-axis.

For part (b),

$y=2f(x-1)$

First, $x-1$ moves the graph 1 unit right. Then the factor 2 stretches the graph vertically by scale factor 2.

Use points from the original graph:

(1,3) to (2,6)

(2,0) to (3,0)

(4,-3) to (5,-6)

(6,0) to (7,0)

(7,3) to (8,6)

Draw the transformed parabola through these points.

Answer: reflection in the x-axis; transformed graph through (2,6), (3,0), (5,-6), (7,0), (8,6)

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(..... ,)
(1)

(ii) $y = f(x) + 7$

(..... ,)
(1)

(iii) $y = f(2x)$

(..... ,)
(1)

(Total for Question 24 is 3 marks)

Worked solution

November 2025 | Question 24 | 3 marks

Family: Write the transformed equation • Variant: Transform an explicit polynomial and simplify

Method

The turning point of $y=f(x)$ is $(4,3)$.

For

$$y=f(x+5)$$

the graph moves 5 units left:

$(4,3)$ to $(-1,3)$

For

$$y=f(x)+7$$

the graph moves 7 units up:

$(4,3)$ to $(4,10)$

For

$$y=f(2x)$$

the x-coordinate is divided by 2:

$(4,3)$ to $(2,3)$

Answer: $(-1,3)$, $(4,10)$, and $(2,3)$